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12 P

Quarterly Examination - 2025

MATHEMATICS

Reg. No. | | | | | | | |

Time : 3.00 hrs. Max. Marks : 90

PART – I

20 x 1 = 20

Answer all the questions.

1. $|adj(adjA)| = |A|^{16}$, then the order of the square matrix A is a) 2 b) 3 c) 5 d) 4
2. If A, B and C are invertible matrices of some order which one of the following is not true?
   a) $adj A = |A|A^{-1}$ b) $adj(AB) = (adjA)(adj B)$ c) $det A^{-1} = (det A)^{-1}$ d) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$
3. If AA^{-1} is symmetric, then $A^2 =$ a) A^{-1} b) $(A^T)^2$ c) A^T d) $(A^{-1})^2$
4. The value of $\left(\frac{1+i}{\sqrt{2}}\right)^8 + \left(\frac{1-i}{\sqrt{2}}\right)^8$ is a) 8 b) 4 c) 2 d) 6
5. If $z = x + iy$ is a complex number such that $|z+2| = |z-2|$, then the locus of z is a) real axis b) imaginary axis c) ellipse d) circle
6. The value of $\sum_{n=1}^{12} i^n$ a) 0 b) 1 c) –1 d) i
7. If α, β and γ are the zeros of $n^3 + pn^2 + qn + r$, then $\sum \frac{1}{\alpha}$ is a) $\frac{q}{r}$ b) $\frac{-q}{r}$ c) $\frac{-q}{p}$ d) $\frac{-p}{r}$
8. A zero of $x^3 + 64$ is a) 0 b) 4 c) 4i d) –4
9. The polynomial $n^3 - kn^2 + 9n$ has three real zeros if k satisfies a) $|k| \le 6$ b) $k = 0$ c) $|k| > 6$ d) $|k| \ge 6$
10. If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, then $\cos^{-1}x + \cos^{-1}y$ is equal to a) $2\frac{\pi}{3}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{6}$ d) π
11. The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x^2-1}$ is a) [1, 2] b) [–1, 1] c) [0, 1] d) [–1, 0]
12. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to a) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ b) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ c) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ d) $\tan^{-1}\left(\frac{1}{2}\right)$
13. The type of conic section for $x^2 - 3 = 5x + 3y$ is a) hyperbola b) ellipse c) circle d) parabola
14. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is a) $\sqrt{ab}$ b) 2ab c) $\frac{a}{b}$ d) ab
15. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$
16. The distance between the planes $x + 2y + 3z + 7 = 0$ and $2x + 4y + 6z + 7 = 0$ is a) $\frac{7}{2\sqrt{2}}$ b) $\frac{\sqrt{7}}{2\sqrt{2}}$ c) $\frac{7}{2}$ d) $\frac{\sqrt{7}}{2}$
17. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, non-zero vectors such that $[\vec{a}, \vec{b}, \vec{c}] = 3$, then $\{[\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}]\}^2$ is equal to
    a) 81 b) 27 c) 9 d) 18
18. If the planes $\vec{r}.[2\hat{i} - \lambda\hat{j} + \hat{k}] = 3$ and $\vec{r}.[4\hat{i} + \hat{j} - \mu\hat{k}] = 5$ are parallel, then the value of λ and μ are
    a) $\frac{1}{2}, -2$ b) $\frac{-1}{2}, 2$ c) $\frac{-1}{2}, -2$ d) $\frac{1}{2}, 2$
19. If $\vec{a}$ and $\vec{b}$ are parallel vectors then $[\vec{a}, \vec{c}, \vec{b}]$ is equal to a) 1 b) 2 c) 0 d) –1
20. If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is a) –1 b) 1 c) 3 d) 9

PART - B

Answer any seven questions. Question number 30 is compulsory. 7 x 2 = 14

21. Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal matrix.
22. If $Adj A = \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$ then find A^{-1}.
23. Find z^{-1}, if $z = (2+3i)(1-i)$
24. Find the value of $\cos^{-1}\left[\cos\left(\frac{7\pi}{6}\right)\right]$.
25. Find the vertices, foci for the hyperbola $9x^2 - 16y^2 = 144$.

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26. Prove that $[\vec{a}-\vec{b},\ \vec{b}-\vec{c},\ \vec{c}-\vec{a}] = 0$

27. Find the distance from a point (2, 5, –3) to the plane $\vec{r}.(6\hat{i}-3\hat{j}+2\hat{k}) = 5$

28. If α and β are the roots of $x^2 + 5x + 6 = 0$, then show that $\alpha^2 + \beta^2 = 13$.

29. Show that the equation $2x^2 - 6x + 7 = 0$ cannot be satisfied by any real values of x.

30. Obtain the equation of the circle for which (3, 4) and (2, –7) are the ends of a diameter.

PART - III

Answer any seven questions. Q.No.40 is compulsory. **7 x 3 = 21**

31. Find the rank of matrix $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$

32. Find the square root of $-6 + 8i$

33. Find a polynomial equation of minimum degree with rational coefficients, having $2i + 3$ as a root.

34. Solve the equation $x^3 - 3x^2 - 33x + 35 = 0$

35. Find the value of $\sin^{-1}\left(\sin\frac{5\pi}{9}\cos\frac{\pi}{9} + \cos\frac{5\pi}{9}\sin\frac{\pi}{9}\right)$

36. Solve $\left(\frac{1-x}{1+x}\right) = \frac{1}{2}\tan^{-1}x$ for $x > 0$

37. Find the equation of the parabola whose vertex is (5, –2) and focus (2, –2)

38. Find the magnitude and the direction cosines of the torque about the point (2, 0, –1) of a force $2\vec{i}+\vec{j}-\vec{k}$, whose line of action passes through the origin.

39. Determine whether the three vectors $2\vec{i}+3\vec{j}+\vec{k}$, $\vec{i}-2\vec{j}+2\vec{k}$ and $3\vec{i}+\vec{j}+3\vec{k}$ are coplanar.

40. The maximum and minimum distances of the Earth from the sun respectively are 152×10^6 km and 94.5×10^6 km. The sun is of one focus of the elliptical orbit. Find the distance from the sun to the other focus.

PART - IV

Note : Answer all the questions. **7 x 5 = 35**

41. a) Solve the following linear equation by Cramer's rule.

$\frac{3}{x}-\frac{4}{y}-\frac{2}{z}-1=0,\ \frac{1}{x}+\frac{2}{y}+\frac{1}{z}-2=0,\ \frac{2}{x}-\frac{5}{y}-\frac{4}{z}+1=0$ **(OR)** b) Solve : $\tan^{-1}\left(\frac{x-1}{x-2}\right)+\tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$

42. a) If $z = x + iy$ and $\arg\left(\frac{z-1}{z+2}\right) = \frac{\pi}{4}$ show that $x^2 + y^2 + 3x - 3y + 2 = 0$ **(OR)** b) Solve : $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$

43. a) Prove by vector method. $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$ **(OR)**

b) Identify the type of conic and find centre, foci, vertices of $18x^2 + 12y^2 - 144x + 48y + 120 = 0$

44. a) A boy is walking along the path $y = ax^2 + bx + c$ through the points (–6, 8), (–2, –12) and (3, 8). He wants to meet his friend at P(7, 60). Will he meet his friend? (Use Gaussian elimination method). **(OR)**

b) Test the consistency of $2x + 2y + z = 5$, $x - y + z = 1$, $3x + y + 2z = 4$ if consistent solve by using rank method.

45. a) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of 4m when it is 6m away from the point of projection. Finally, it reaches the ground 12m away from starting point. Find the angle of projection.

(OR) b) Find the value of $\cos^{-1}\left[\cos\left(\frac{4\pi}{3}\right)\right] + \cos^{-1}\left[\cos\left(\frac{5\pi}{3}\right)\right]$

46. a) Find all cube roots of $\sqrt{3} + i$ **(OR)**

b) Find the non-parametric form of a vector equation and Cartesian equation of the plane passing through the point (1, –2, 4) and perpendicular to the plane $x + 2y - 3z = 11$ and parallel to the line $\frac{x+7}{3} = \frac{y+3}{-1} = \frac{z}{1}$. Solve the equation.

47. a) $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ If it is known that $\frac{1}{3}$ is a solution. **(OR)**

b) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ and $0 < x, y, z < 1$ then show that $x^2 + y^2 + z^2 + 2xyz = 1$

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